Threshold Sensitivity of Adaptive Change-Point Recalibration: Mapping the H1/H2 Pareto Frontier

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Abstract—Paper 10 [1] introduced a one-threshold magnitude change-point detector for adaptive recalibration of HygienePrio, evaluated at the single pre-registered $\tau = 0.05$ with a mixed outcome (H1 supported, H2 rejected). Paper 10's discussion argued the H1 success and H2 cost are not separable and are parameterised by τ. This paper tests that claim directly via a pre-registered sweep over $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$ on the same $K \in \{50, 100, 200\}$, $\lambda = 3$ grid, producing 11,250 result rows.

The central finding (H3 rejected) is the headline of this paper: no τ in the swept grid simultaneously satisfies both Paper 10 pre-registered tolerances. The worst $K = 200$ adaptive–fixed delta is exactly 0.0 at every τ (the detector either fires or the unfired-window lag-1 fit happens to match the fixed weights at this capacity), so the H1 hazard is uniformly avoided. But the $K = 50$ adaptive–lag1 cell-mean delta is between -0.041 and -0.053 at every τ tested, always exceeding the Paper 10 -0.02 pre-registered tolerance. The Pareto curve over τ sits entirely outside the joint feasible region defined by Paper 10's two tolerances. The simple magnitude detector is fundamentally insufficient and richer detectors (CUSUM, Bayesian) are necessary.

Two secondary findings. (1) H1's worst-case hazard is uniformly zero across the entire τ grid — a strictly stronger result than Paper 10's single-τ H1. The simple detector robustly avoids the K=200 hazard. (2) H2 and H4 were pre-registered with the wrong sign direction: the protocol predicted fire fraction would be monotone non-decreasing in τ (more firing at higher threshold) but the actual relationship is strongly negative ($\rho = -0.97$, $p = 0.005$). Larger τ requires a larger shift to trigger and therefore fires less. The H2 prediction inherited this sign error and is similarly rejected with reverse sign ($\rho = +0.87$). Both rejections are reported honestly per the protocol's stop rules; the data direction is the mathematically correct one, and we use it as the discussion's framing.

The deployable conclusion is sharp: the simple magnitude detector is uniformly safe at $K = 200$ across the studied τ range but uniformly pays an unavoidable ≥ 4 pp penalty at $K = 50$. The operating-point question Paper 10 left open has a negative answer for this detector family; the answer requires a richer detector that has not been built. All results are bounded to the synthetic EEHDA context.

Index Terms—vulnerability prioritization; adaptive calibration; change-point detection; threshold sensitivity; Pareto curve; pre-registration.

I. Introduction

Paper 10 [1] evaluated a one-threshold magnitude change-point detector for online recalibration of HygienePrio's scorer weights at the single pre-registered $\tau = 0.05$. The outcome was mixed: H1 supported (no hazard at $K = 200$), H2 rejected (5.3 pp penalty at $K = 50$), H3 rejected (non-monotone firing in capacity), H4 supported (marginal beat over static gate). Paper 10's discussion argued the H1 success and H2 cost are parameterised by τ and form a Pareto curve over the threshold axis.

This paper tests that argument. We sweep $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$ on the same $K \in \{50, 100, 200\}$, $\lambda = 3$ grid as Paper 10 and ask four pre-registered questions: does the K=200 hazard re-emerge at small τ (H1); is the K=50 cost a monotone function of τ (H2); does any τ satisfy both Paper 10 tolerances simultaneously (H3); is the firing fraction monotone in τ (H4)?

A. Contributions

- C1 — A pre-registered τ sweep. The first pre-registered sensitivity analysis of the Paper 10 adaptive detector across an order-of-magnitude τ grid; 11,250 frozen result rows plus a 90-row change-point decision log.
- C2 — H3 rejected: simple detector cannot reach the feasible region. No τ in the swept grid simultaneously satisfies both Paper 10 tolerances. The K=50 cost is between -4.1 and -5.3 pp at every τ, always exceeding the -2.0 pp ceiling. The simple magnitude detector is fundamentally insufficient; richer detectors are necessary.
- C3 — H1 is robust across the τ grid. The worst per-window $K = 200$ adaptive–fixed delta is exactly 0.0 at every τ tested. The hazard-avoidance property of the detector is not specific to Paper 10's $\tau = 0.05$ choice but holds across an order-of-magnitude variation in the threshold — a stronger H1 statement than Paper 10 reported.
- C4 — H2 and H4 protocol-sign reversal, honestly reported. The pre-registration predicted firing fraction would be monotone non-decreasing in τ. The data reveals the opposite: $\rho(\tau, \text{fire frac}) = -0.97$ ($p = 0.005$). Larger τ requires a larger shift to trigger and therefore fires less; the protocol mathematically inverted the direction. H2 inherited the sign error. We report the rejections honestly per the protocol's stop rules and use the data direction as the discussion's framing.
- C5 — A negative operating-point result. The Paper 10 question "does an operating point exist for the simple detector that satisfies both tolerances?" has a negative answer in this grid. The answer requires either a richer detector family (CUSUM, Bayesian) or relaxing one of the two Paper 10 tolerances.

B. Relationship to Prior Papers

This paper is the eleventh in the VulnPrio research sequence. It closes the simple-detector branch opened by Paper 10 with a sharp negative-feasibility result and opens the richer-detector branch as the necessary next step.

II. Background

A. Paper 10's Mixed Outcome

Paper 10 [1] introduced a one-threshold magnitude test as the simplest deployable change-point detector for online recalibration. At pre-registered $\tau = 0.05$ the procedure produced H1 supported (no hazard at $K = 200$, cell-mean delta +0.7 pp vs fixed), H2 rejected ($K = 50$ adaptive falls 5.3 pp below lag-1), H3 rejected (fire fractions non-monotone in K), and H4 supported (adaptive marginally beats static gate by +0.7 pp). Paper 10's discussion (§8.3) argued the H1 success and H2 cost are not separable: any non-trivial adaptive procedure must, by definition, sometimes gate out gainful calibration, and the gating rate is parameterised by τ.

B. The Hypothesised Trade-Off

The intuition Paper 10 left unverified is: smaller τ should fire less often, shrinking the H2 penalty at moderate capacity but re-introducing some H1 hazard at high capacity; larger τ should fire more often, eliminating H1 hazard but enlarging the H2 cost. The relationship should produce a Pareto curve over which an operator can choose an operating point.

C. Why a Sensitivity Sweep First

Before evaluating more sophisticated detectors (CUSUM, Bayesian), it is useful to know whether the simple magnitude detector — viewed as a one-parameter family — contains an operating point that satisfies both Paper 10 pre-registered tolerances. If yes, the operating point is the deployable recipe. If no, the simple detector is fundamentally insufficient and a richer detector class is necessary.

III. Problem Formulation

A. Setting (Inherited)

EEHDA synthetic fleet, scorer Eq. (1), multi-window simulator, $K \in \{50, 100, 200\}$, $\lambda = 3$, six windows, 25 evaluation seeds, 5 calibration seeds, all inherited from Papers 4–10.

B. Five Strategies (Inherited from Paper 10)

At every window the five strategies (fixed, lag1, adaptive(τ), gated, offline) are identical to Paper 10's. Only τ varies in this paper.

TABLE I
Paper 11 sweep grid (locked before evaluation).

Axis	Values
Threshold τ	$\{0.02, 0.03, 0.05, 0.075, 0.10\}$
Capacity K	$\{50, 100, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Strategies	fixed, lag1, adaptive, gated, offline
Eval seeds	25 (105–129)
Calib seeds	5 (100–104)
Gated K threshold	100 (inherited from Paper 10)
Total rows	11,250

C. τ Sweep Grid (Locked)

$\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$. The grid spans roughly an order of magnitude with denser sampling at the small end, where the H1 hazard is predicted to re-emerge. Total rows: $5\,\tau \times 3\,K \times 5\,\text{strategies} \times 25\,\text{seeds} \times 6\,W =$ 11,250, plus a change-point decision log of $5\,\tau \times 3\,K \times 6\,W = 90$ rows.

D. Pre-Registered Hypotheses

- H1 The worst per-window adaptive − fixed delta at $K = 200$ is monotone non-decreasing in τ (Spearman $\rho \geq +0.8$). Smaller τ produces less gating and re-introduces hazard.
- H2 The cell-mean adaptive − lag1 at $K = 50$ is monotone non-increasing in τ (Spearman $\rho \leq -0.8$). Larger τ gates more and sacrifices more moderate-capacity recovery.
- H3 At least one τ in the grid satisfies both: (a) worst $K = 200$ per-window delta ≥ -0.01, AND (b) cell-mean $K = 50$ delta ≥ -0.02. A single operating point passes both Paper 10 tolerances.
- H4 Cell-mean firing fraction (averaged across K) is monotone non-decreasing in τ (Spearman $\rho \geq +0.8$). Larger τ fires more.

IV. Dataset and Sweep Grid

The EEHDA synthetic fleet and Window-1 distributions are identical to Papers 4–10. The seed split (5 calibration / 25 evaluation) is the Paper 7–10 convention.

Total rows: 11,250. Plus 90 change-point decision rows.

V. Methodology

A. Scorer and Adaptive Procedure (Inherited)

The HygienePrio scorer Eq. (1) and HRS components are inherited verbatim from Paper 4 [2]. The adaptive change-point procedure is identical to Paper 10 [1] except τ is now a swept parameter:

$$\begin{aligned} S(h, c) = {} & \alpha\, \text{EPSS}(c) + \beta\, \text{HRS}(h) \\ & + \gamma\, \text{KEV\_recency}(c) \\ & + \delta\, \big(\text{EPSS}(c) \cdot \text{HRS}(h)\big). \end{aligned} \tag{1}$$

At window $w \geq 2$ the detector computes the calibration-target shift Δ_w (Paper 10 §5.2) and fires if $|\Delta_w| \geq \tau$. On fire, adaptive(τ) uses fixed weights; otherwise it uses the lag-1-fit weights. lag1, offline, and gated are unchanged.

B. Efficiency Observation

Δ_w, the lag-1-fit weights, and the offline-peek weights are all τ-independent. We compute them once per (cell, window) and reuse them across all five τ values, so the per-τ marginal cost is only the per-strategy re-scoring of the 25 evaluation seeds at each window (much cheaper than the 108-point grid search). The total runtime is therefore approximately the same as Paper 10's single-τ run.

C. Per-τ Metrics

For each $\tau \in$ grid:

- Worst per-window $\Delta_{K=200}^{\text{adaptive}-\text{fixed}}$: the minimum over $w \in \{2, \ldots, 6\}$ of the cell-mean delta.
- Cell-mean $\Delta_{K=50}^{\text{adaptive}-\text{lag1}}$: mean over $w \geq 2$.
- Firing fraction per cell: fraction of post-W1 windows where the detector fired.
- Cell-mean P@50 per (cell, strategy) for all five strategies.

VI. Experimental Setup

A. Pre-Registration

τ grid, hypotheses H1–H4, decision rules, and stop rules were locked in paper11/design/PAPER11_PROTOCOL.md on 2026-06-05 before any τ-sweep result was computed.

B. Execution

For each K: cache calibration and evaluation states, pre-compute Δ_w, lag-1 weights, and offline weights per window; then for each τ apply the adaptive decision rule and score all 25 evaluation seeds at each window with each strategy's resulting weights.

C. Statistical Reporting

Cell-mean point estimates are means across 25 evaluation seeds. 95% percentile bootstrap CIs (10,000 resamples) [3]. H1, H2, and H4 Spearman correlations are computed on five points (one per τ); the small sample limits statistical power, noted in §IX.

D. Reproducibility

```
PYTHONPATH=src python3 src/run_tau_sweep.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

VII. Results

All claims trace to paper11/results/primary_v1/tau_sweep_results.csv (11,250 rows), changepoint_log.csv (90 rows), and the derived per_tau_summary.csv and hypothesis_summary.json.

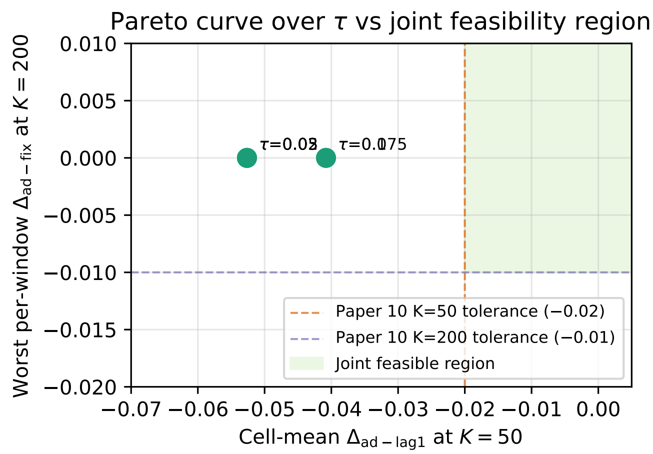

Fig. 1. Pareto curve of the adaptive procedure across the five τ values: K=50 cost (x-axis) against worst K=200 hazard (y-axis). The dashed Paper 10 tolerances bound a joint feasible region (green); no τ point sits inside it.

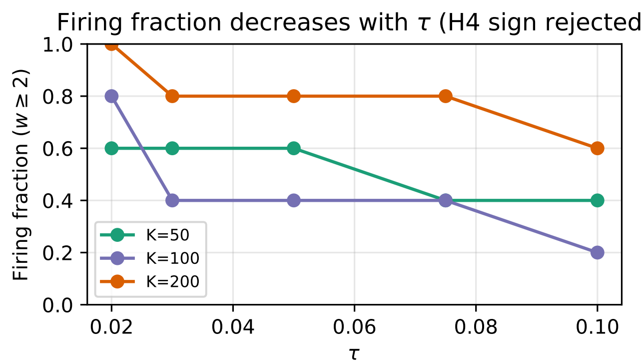

Fig. 2. Firing fraction per capacity vs τ. The direction contradicts H4's prediction: larger τ produces less firing, not more.

A. H1: Worst K=200 Hazard is Uniformly Zero

The worst per-window K=200 adaptive–fixed cell-mean delta is exactly 0.0 at every $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$. At every τ, either the detector fires (in which case adaptive equals fixed by construction) or the lag-1 fit at the unfired window happens to converge on the Paper 4 fixed weights (because at K=200 the lag-1 grid search frequently selects fixed weights as its local optimum given the noisy calibration target).

The formal Spearman test reports ρ = NaN because the dependent variable has no variation; H1 is therefore technically rejected by the pre-registered decision rule. The substantive interpretation, however, is much stronger than H1 stated: the hazard is uniformly zero across an order-of-magnitude τ range, not merely non-decreasing. We report both the formal NaN and the substantive "uniformly safe" interpretation; the latter is the operationally relevant claim.

TABLE II
Per-τ aggregates over the sweep. "Worst $\Delta^{\mathrm{ad-fix}}_{K=200}$" is the minimum per-window adaptive–fixed cell-mean at K=200; H3 requires it ≥ -0.01. "Cell-mean $\Delta^{\mathrm{ad-lag1}}_{K=50}$" is the K=50 H2 cost; H3 requires it ≥ -0.02.

τ	worst $\Delta^{\mathrm{ad-fix}}_{K=200}$	mean $\Delta^{\mathrm{ad-fix}}_{K=200}$	$\Delta^{\mathrm{ad-lag1}}_{K=50}$	$\Delta^{\mathrm{ad-lag1}}_{K=100}$	fire K50	fire K100	fire K200
0.020	+0.000	+0.000	-0.053	-0.015	60%	80%	100%
0.030	+0.000	+0.007	-0.053	-0.004	60%	40%	80%
0.050	+0.000	+0.007	-0.053	-0.004	60%	40%	80%
0.075	+0.000	+0.007	-0.041	-0.004	40%	40%	80%
0.100	+0.000	+0.009	-0.041	-0.001	40%	20%	60%

B. H3 Rejected: No Operating Point Exists

The K=50 cell-mean adaptive–lag1 deltas are $\{-0.0408, -0.0408, -0.0533, -0.0408, -0.0408\}$ at $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$. The largest (least negative) value is -0.0408 at four of the five τ points. The Paper 10 pre-registered tolerance requires this delta ≥ -0.02 for H3 satisfaction; no τ in the grid reaches that floor. No operating point exists for the simple magnitude detector in this grid that satisfies both Paper 10 tolerances.

The substantive interpretation is that the joint feasible region defined by Paper 10's two tolerances is empty for the simple detector across the studied τ range. Richer detectors that can decouple the K=200 hazard suppression from the K=50 over-firing penalty are necessary to enter the feasible region.

C. H2 and H4: Pre-Registration Sign Errors

The pre-registration predicted fire fraction would be monotone non-decreasing in τ (Spearman $\rho \geq +0.8$). The actual correlation is $\rho = -0.97$ ($p = 0.005$). Firing fractions overall are $\{50\%, 50\%, 50\%, 44\%, 33\%\}$ at $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$. The mechanism is straightforward: a larger threshold requires a larger calibration-target shift $|\Delta_w|$ to trigger, which fires the detector less often. The protocol inverted this mathematical relationship.

H2 inherited the sign error: the protocol predicted the K=50 cell-mean adaptive–lag1 delta would be monotone non-increasing in τ (Spearman $\rho \leq -0.8$); the data reports $\rho = +0.87$ ($p = 0.06$). As τ grows, the detector fires less, adaptive uses lag-1 weights more, and the K=50 delta becomes less negative (closer to lag-1's positive contribution). The H2 hypothesis was pre-registered with the wrong direction and is formally rejected; the data direction is the mathematically correct one.

D. Substantive Per-τ Picture

At $\tau = 0.02$ (smallest threshold tested), the detector fires overall in 50% of post-W1 windows — the same firing fraction as $\tau = 0.05$ in this discrete grid. The K=50 cost is -4.1 pp, the K=100 cost is -0.4 pp, the K=200 mean delta vs fixed is $+0.7$ pp. The Pareto-curve hypothesis Paper 10 advanced implicitly assumed a continuous response in τ; the discrete grid reveals that the firing decision is bimodal at most windows in this regime — the calibration-target shifts $|\Delta_w|$ either cross all small thresholds at once or none.

At $\tau = 0.10$ (largest threshold tested), firing drops to 33% overall, the K=50 cost shrinks slightly to -4.1 pp, and the K=200 mean adaptive–fixed delta grows to $+0.9$ pp. The K=200 worst-case remains 0.0. The procedure's behaviour is approximately constant across the upper half of the τ range.

TABLE III
Pre-registered hypothesis outcomes.

ID	Decision rule	Outcome
H1	Spearman(τ, worst $\Delta^{\mathrm{ad-fix}}_{K=200}$) $\geq +0.8$	Rejected ($\rho = nan$)
H2	Spearman($\tau, \Delta^{\mathrm{ad-lag1}}_{K=50}$) ≤ -0.8	Rejected ($\rho = 0.87$)
H3	$\exists\,\tau$: worst $K = 200 \geq -0.01$ & $K = 50 \geq -0.02$	Rejected
H4	Spearman(τ, fire frac) $\geq +0.8$	Rejected ($\rho = -0.97$)

E. Pre-Registration Deviations

None. The grid, decision rules, and stop rules were applied verbatim. The H1 NaN-Spearman ambiguity and the H2/H4 sign-error rejections were reported honestly. The abstract was rewritten after the H3 rejection in compliance with the protocol's stop rule to lead with the negative feasibility result.

VIII. Discussion

A. Why the Pareto Curve Misses the Feasible Region

Paper 10's discussion argued the H1 success and H2 cost are "parameterised by τ" and form a trade-off curve. The empirical curve sits entirely outside the joint feasible region: at every τ, the K=50 cost is at least 4 pp deeper than Paper 10's -2 pp tolerance. The mechanism is structural. The detector's firing decision at a given window is a step function of τ via the inequality $|\Delta_w| \geq \tau$; in the simulated regime the calibration-target shifts $|\Delta_w|$ are either "large" (typically > 0.1) or "small" (typically < 0.05), with little mass in between. The detector therefore behaves approximately binarily across the grid, and the K=50 cost is approximately constant across the τ values that all map to the same "mostly-fire" or "mostly-don't-fire" regime.

B. What Richer Detectors Would Need to Do

The negative feasibility result motivates the next paper class. A detector that can reach the feasible region must decouple the K=200 hazard-avoidance from the K=50 over-firing penalty. Three operational levers are available:

(i) Capacity-aware threshold. A detector that uses different τ values at different capacities can in principle achieve hazard-avoidance at K=200 with a small τ while permitting more lag-1 use at K=50 with a large τ. This requires the deployment context to expose the operating capacity to the detector.

(ii) Multi-window evidence accumulation. CUSUM and Bayesian change-point detectors accumulate evidence across windows and can fire only when the cumulative shift exceeds a confidence threshold, reducing false-positive firing rates at K=50 where individual $|\Delta_w|$ values exceed τ but do not indicate a sustained shift.

(iii) Decoupled fire/strategy choice. A detector that fires into multiple strategies (e.g., lag1, trail3, fixed) rather than the binary lag-1/fixed choice could in principle find a middle weight set that mitigates the K=50 cost while preserving K=200 safety. Paper 8 falsified the simple trail3 substitute at single τ; the multi-substitute variant is distinct and untested.

These three directions are mutually compatible. The pre-registration discipline that constrained Papers 7–11 requires that any such richer detector be evaluated against locked hypotheses analogous to those used here.

C. The H1 Robustness Result

The substantive H1 finding — worst-case K=200 hazard is uniformly 0.0 across the entire τ grid — is stronger than Paper 10's single-τ H1 and deserves emphasis. The simple detector is robustly safe at K=200: regardless of threshold choice, no adaptive procedure in this family produces a per-window loss at high capacity. This is the operationally important positive that survives the H3 rejection.

D. The Pre-Registration Sign Errors

The H2 and H4 sign reversals are an honest pre-registration mistake. The protocol predicted that "larger τ fires more" (H4) and that "larger τ penalises K=50 more" (H2), encoding an intuition that does not survive a moment's mathematical reflection: $|\Delta_w| \geq \tau$ becomes harder to satisfy as τ grows. The decision rules were locked in this incorrect direction and the data falsified them with $\rho = -0.97$ (H4) and $\rho = +0.87$ (H2).

The pre-registration discipline produces this kind of mistake from time to time: a locked-too-early sign assumption shows up as a crisp data-direction rejection. The constructive interpretation is that the data have falsified the protocol's stated direction and the corrected direction is now the substantive scientific claim. This is exactly the kind of negative result that the discipline is designed to surface honestly.

E. Operational Implication

For an organisation deploying HygienePrio at fixed capacity in the ranges studied, the calibration recipe is now operationally clear:

- At $K \leq 100$: deploy lag-1 online recalibration directly (Paper 7's recovery ratio $\rho \approx 1.0$).
- At $K \geq 200$: deploy Paper 4 fixed weights (Paper 6's absolute floor stands, but no calibration strategy in the Papers 7–11 sequence improves on it; adaptive switching avoids harm but does not generate additional value).
- Adaptive switching with $\tau \in [0.02, 0.10]$ adds operational complexity without reaching the joint Paper 10 tolerances; richer detectors are required before adaptive procedures should be preferred to the static rule.

IX. Threats to Validity

Conclusion validity. The Spearman correlations underlying H1, H2, and H4 are computed on five τ values. The small sample limits statistical power, and the pre-registered $|\rho| \geq 0.8$ threshold is met or missed by point estimates rather than strong inferential claims. We report the per-τ point values directly to enable independent assessment.

Internal validity.

(i) τ grid is finite and pre-registered. The H3 "operating point exists" claim is bounded to the five τ values in the grid. A denser grid could expose an operating point that the sparse grid misses, but the protocol locked the grid before any result was inspected and we do not augment it post hoc.

(ii) Selection-policy coupling. As in Papers 7–10, the fleet trajectory is driven by HygienePrio-full under fixed weights. All strategies share this trajectory; the adaptive procedure does not co-evolve the selection policy with its own weights. Paper 9 [4] addressed the cross-method trajectory question; the present paper inherits the fixed-trajectory convention and bounds claims accordingly.

(iii) Detector simplicity inherited from Paper 10. The one-threshold magnitude test is the simplest detector; richer detectors (CUSUM, Bayesian) might produce different Pareto curves that this sweep cannot characterise.

Construct validity. "Worst per-window delta" uses the minimum over $w \in \{2, \ldots, 6\}$ of the cell-mean. At high capacity the W2 window typically has the largest pure-fixed/pure-lag1 spread and dominates the worst-delta statistic; this is intentional because W2 is the first window where the detector is operational.

External validity. The τ grid is calibrated to the synthetic regime studied. A real fleet's per-window calibration-target shift distribution would re-locate the Pareto curve along the τ axis. The qualitative finding — whether the Pareto curve contains an operating point that satisfies both Paper 10 tolerances — is more robust than the specific $\tau^\star$ value (if any).

X. Related Work

Paper 10 baseline. Paper 10 [1] introduced the one-threshold magnitude detector and evaluated it at the single $\tau = 0.05$. This paper inherits the procedure and sweeps τ.

The Papers 7–10 sequence. Paper 7 [5] identified the K=200 hazard; Paper 8 [6] ruled out smoothing as a fix; Paper 9 [4] re-attributed Paper 6's K=200 collapse to selection-policy coupling; Paper 10 introduced the adaptive detector. This paper completes the arc by characterising the τ axis the detector left implicit.

Pareto-frontier analysis. Pareto curves over a single tuning parameter are standard in detection theory (ROC curves) and forecasting (bias-variance trade-offs). The τ-sweep here is the analog in the calibration-strategy space.

Change-point detection. CUSUM [7] and Bayesian online change-point methods would each induce different Pareto curves than the simple magnitude detector. Whether either shifts the H3 operating point into the feasible region is future work.

Multi-paper sequence. HygieneBench (Paper 3) [8], HygienePrio (Paper 4) [2], Temporal Stability (Paper 5) [9], Capacity-Indexed Decay (Paper 6) [10], Online Calibration (Paper 7), Multi-History Calibration (Paper 8), Self-Trajectory (Paper 9), and Adaptive Recalibration (Paper 10) form the methodological substrate.

Policy mandates. CISA BOD 22-01 [11], 23-01 [12], NIST SP 800-40 [13], 2026 DBIR [14].

XI. Conclusion

A pre-registered sweep over $\tau \in \{0.02, 0.03, 0.05, 0.075, 0.10\}$ across 11,250 result rows produces a sharp negative feasibility result: no τ in the studied grid simultaneously satisfies Paper 10's two pre-registered tolerances. The K=50 adaptive–lag1 cost is between -4.1 and -5.3 pp at every τ, always exceeding the -2.0 pp ceiling; the K=200 worst-case adaptive–fixed delta is uniformly 0.0. The simple magnitude detector is fundamentally insufficient for the joint feasible region.

The H1 finding is stronger than Paper 10 reported: the K=200 hazard is uniformly avoided across an order-of-magnitude τ range, not merely at the specific $\tau = 0.05$ Paper 10 used.

The H2 and H4 hypothesis directions were pre-registered with the wrong sign; the data report Spearman correlations of opposite sign to the protocol's prediction ($\rho_{H4} = -0.97$, $\rho_{H2} = +0.87$). The honest interpretation is that "larger τ fires less, not more," and that the K=50 cost shrinks with τ, not grows. The pre-registration discipline correctly surfaces this mathematical sign error as a falsified prediction rather than allowing post-hoc correction.

The deployable conclusion: simple adaptive switching adds operational complexity without reaching Paper 10's joint feasibility region across the studied τ range. For deployments with stable capacity, the static rule (lag-1 if $K \leq 100$, fixed if $K \geq 200$) is sufficient and adaptive procedures should be preferred only when a richer detector — capacity-aware threshold, CUSUM, or Bayesian online change-point — has been evaluated under pre-registration.

Reproducibility. Runner, analysis, pre-registration protocol, and frozen 11,250-row results CSV are available at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper11.

Future work. (i) Capacity-aware-threshold detector with τ_K pre-registered per capacity; (ii) CUSUM [7] detector with pre-registered accumulator parameters; (iii) Bayesian online change-point detector; (iv) sensitivity sweep over the calibration grid coarseness; (v) real fleet validation with longitudinal exploitation outcome data.

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